



National
Qualifications
2024

2024 Biology

Higher - Paper 1

Question Paper Finalised Marking Instructions

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Marking instructions for each question

Question	Response	Mark
1.	D	1
2.	B	1
3.	D	1
4.	D	1
5.	B	1
6.	C	1
7.	B	1
8.	B	1
9.	A	1
10.	D	1
11.	B	1
12.	B	1
13.	A	1
14.	C	1
15.	C	1
16.	B	1
17.	D	1
18.	A	1
19.	C	1
20.	C	1
21.	C	1
22.	A	1
23.	D	1
24.	A	1
25.	C	1

[END OF MARKING INSTRUCTIONS]



National
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2024 Biology

Higher - Paper 2

Question Paper Finalised Marking Instructions

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Marking Instructions for each question

Question			Expected response	Max mark	Additional guidance
1.	(a)		Translation ribosome (1) (1)	2	NOT cytoplasm
	(b)	(i)	1800	1	
		(ii)	alternative (RNA) splicing	1	NOT splicing
	(c)	(i)	his/histidine	1	
		(ii)	(results in a premature) stop codon (1) protein would be shorter OR protein would contain fewer amino acids (1)	2	NOT short protein NOT non-functional or different protein
2.	(a)		3187.5	1	
	(b)	(i)	deletion (1) duplication (1)	2	
		(ii)	protein is not made/missing OR gene is not expressed OR gene that codes for protein is removed	1	NOT non-functional/different protein is made
	(c)	(i)	gene/section of chromosome is added from/to homologous chromosome	1	
		(ii)	beneficial/advantageous mutations can occur in one copy of the gene (1) while the other copy of the gene can still be expressed OR while the other copy of the gene can still code for protein (1)	2	

Question			Expected response	Max mark	Additional guidance
3.	(a)	(i)	Phylogenetics	1	
		(ii)	fossils (1) sequence data (1)	2	
	(b)	(i)	computer/statistical	1	
		(ii)	they have a more recent common ancestor OR they diverged more recently OR there are fewer differences in their base sequences	1	
4	(a)		citric acid cycle	1	
	(b)	(i)	induced fit	1	
		(ii)	fumarate leaves so succinate can bind to the active site	1	
	(c)		Type of inhibition: Competitive (1) Justification: malonate/it has a similar shape/structure to succinate OR malonate/it is complementary to the active site (1)	2	NOT malonate has a similar shape to the active site

Question			Expected response	Max mark	Additional guidance
5.	(a)		concentration/mass of glucose in media OR type/age/surface area of muscle/tissue	1	NOT amount/quantity of glucose/muscle/tissue
	(b)	(i)	2.3/2.33/2.333... / $2\frac{1}{3}$	1	
		(ii)	42	1	
	(c)		• good supply of glucose/oxygen • for respiration • provides ATP/energy required for temperature regulation/homeostasis (Any 2)	2	
	(d)		shivering (1) muscle contraction generates heat (1) OR vasoconstriction/narrowing of blood vessels (1) less blood flow to skin so less heat is lost (1) OR hair erector muscles contract/hair raises (1) traps an insulating layer of air (1)	2	NOT blood vessels move away from the skin

Question			Expected response	Max mark	Additional guidance
6.	(a)	(i)	Any value from 5 to 5.2	1	
		(ii)	decreased heart/breathing/metabolic rate OR decreased oxygen consumption/CO ₂ production/respiration/metabolic rate	1	NOT low heart rate etc... NOT decreased activity
	(b)	(i)	5.5 OR 6	1	
		(ii)	Type of dormancy: predictive (1) Reason: began before the decrease in (air) temperature OR began before the onset of adverse conditions (1)	2	
		(iii)	saves/conserves energy OR survive when metabolic costs would be too high	1	NOT avoids adverse conditions
7.	(a)		150	1	
	(b)	(i)	From 0 to 175 minutes it increases from 0 to 45 g/l (1) From 175 minutes (to 250) it remains constant/levels off (1)	2	Award one mark only for increases to 175 then levels off To achieve 2 marks both units must be given at least once.
		(ii)	100	1	
	(c)		ethanol concentration is 0(g/l) OR all ethanol has been used up	1	NOT no nutrients left
	(d)		40	1	
8.	(a)	(i)	vector	1	
		(ii)	Allows the plasmid to replicate/make copies of itself	1	
	(b)		restriction endonuclease	1	
	(c)		Gene: antibiotic resistance (1) Explanation: transformed bacteria/they will grow/survive in the presence of antibiotic (1)	2	Accept other correct examples e.g. fluorescence gene

Question			Expected response	Max mark	Additional guidance
9.	A		1. complete double circulatory system 2. two atria and two ventricles 3. no mixing of oxygenated and deoxygenated blood 4. blood is (pumped out) at high pressure 5. efficient oxygen delivery to cells/tissues/organs 6. enables/supports high metabolic rates (Any 4)	4	NOT no mixing of blood
	B		1. occurs in the absence of oxygen 2. takes place in the cytoplasm 3. glucose broken down/converted to pyruvate 4. pyruvate is converted to ethanol and CO ₂ 5. pyruvate to ethanol and carbon dioxide is irreversible 6. less ATP produced than aerobic respiration OR ATP produced OR net gain of 2 ATP (Any 4)	4	

Question			Expected response	Max mark	Additional guidance
10.	(a)	(i)	rate of photolysis	1	NOT absorbance
		(ii)	so that no other/external/sun light affects the experiment/algae OR so that only the coloured light affects the experiment/algae	1	NOT so no light affects the experiment
	(b)		axes correctly labelled and scale correct (1) points correctly plotted and joined (1)	2	
	(c)		blue light results in the highest rate of photolysis OR photolysis is fastest with blue light	1	NOT rate of photosynthesis
	(d)		green light is not absorbed/less green light is absorbed OR green light is reflected/transmitted (1) fewer/few/no hydrogen ions (to decolourise DCPIP) OR less/no/little photolysis/photosynthesis (1)	2	

Question			Expected response	Max mark	Additional guidance
11.	(a)	(i)	less photosynthesis OR less light absorbed for photosynthesis (1) less energy/ATP/glucose for growth/fruit production (1)	2	Award 1 mark for: insects spread diseases which damage the plants/strawberries when no other mark is awarded NOT insects cause disease
		(ii)	(Introduce) a predator/pathogen/parasite (of the pest) OR it would prey on/eat/infect the insects/pest	1	
		(iii)	it could harm/prey on/compete with other/non-target species OR it could become invasive	1	
		(iv)	chemical and biological control OR chemical and cultural control OR cultural and biological control OR chemical, biological and cultural control	1	NOT examples
	(b)	(i)	open ends/air flow/ventilation reduce humidity	1	
		(ii)	(prevention is) more effective than treating diseased crops OR decreases use of fungicide/pesticide/chemicals OR less harmful chemicals in the environment	1	
	(c)	(i)	increased/high yield/vigour/growth rate	1	
		(ii)	too much variation in F ₂	1	

Question			Expected response	Max mark	Additional guidance
12.	(a)		naturalised	1	
	(b)		(an introduced species which) spreads rapidly and eliminates native species	1	
	(c)		no/less/few predators/competitors/pathogens/parasites that were found in their native/original habitat/Central America	1	
	(d)	(i)	95	1	
		(ii)	native frogs: decrease close to the lake OR increases further away from the lake OR overall numbers decrease (1) cane toads: increase close to the lake OR increased more closer to the lake than further away from the lake OR overall numbers increase (1)	2	If values are used they must be correct.

Question			Expected response	Max mark	Additional guidance
13.	(a)	(i)	2:7	1	
		(ii)	A - More time spent in misdirected behaviour	1	NOT lots of time spent in misdirected behaviour NOT more misdirected behaviour
	(b)		More (pigs) in each group/farm OR more groups (of pigs) in each farm	1	NOT more pigs/groups NOT repeat and calculate an average NOT more farms
	(c)		failure in sexual/parental behaviour OR stereotypy OR apathy/hysteria/altered levels of activity	1	NOT examples
	(d)		reduced cost/cheaper/increased profit/less land required/less labour intensive OR more cost effective	1	NOT low cost/cheap etc...
	(e)		energy is lost at each/between trophic level/levels of food chain (1) fewer trophic levels/levels of food chain with cereal/crop production (1)	2	

Question			Expected response	Max mark	Additional guidance
14.	(a)	(i)	353 600	1	
		(ii)	can kill/catch larger prey OR less energy used per individual OR more successful hunts	1	NOT can hunt larger prey NOT can kill/catch large prey
	(b)	(i)	rank (order) (1) (with) dominant and subordinates (1)	2	
		(ii)	ritualistic (display) OR appeasement	1	NOT examples
		(iii)	increases the chance of dominant animal's favourable/advantageous genes being passed on to offspring	1	

Question			Expected response	Max mark	Additional guidance
15	A		1. occurs before cell division 2. DNA unwinds/untwists 3. hydrogen bonds between bases/strands break OR hydrogen bonds break separating the strands/unzipping the DNA 4. primers join to (3' end of) template DNA OR primer provides a start point for DNA replication/ DNA polymerase 5. primer is a short strand of nucleotides 6. DNA polymerase adds nucleotides to 3' end of the primer/new strand 7. complementary base pairing occurs OR A - T and C - G 8. sugar of one nucleotide joins phosphate of next OR sugar phosphate backbone forms 9. leading strand is replicated continuously 10. lagging strand is replicated in fragments/discontinuously 11. (DNA) ligase joins fragments 12. two identical DNA molecules produced. Any 8	8	

Question			Expected response	Max mark	Additional guidance
15.	B		1. population/species is split/divided/separated (by an isolation barrier/mechanism) 2. isolation barriers can be geographical, ecological or behavioural 3. prevents one population breeding with the other population OR prevents populations interbreeding OR prevents gene flow between populations 4. geographical leads to allopatric speciation 5. ecological/behavioural leads to sympatric speciation 6. different mutations occur on each side of barrier/in each population 7. some mutations may be beneficial/advantageous 8. different selection pressures exist in each population 9. natural selection occurs 10. some individuals survive and pass on favourable alleles/genes to offspring OR (natural selection is) the non-random increase in frequency of DNA sequences/alleles that increase survival 11. populations can no longer breed with each other/ interbreed to produce fertile young 12. (this results in) two/different/ new species Any 8	8	All 3 barriers required NOT prevents populations breeding NOT populations can no longer breed to produce fertile young

[END OF MARKING INSTRUCTIONS]

General marking principles for Higher Biology

Always apply these general principles. Use them in conjunction with the marking instructions, which identify the key features required in candidates' responses.

- (a) Always use positive marking. This means candidates accumulate marks for the demonstration of relevant skills, knowledge and understanding; marks are not deducted for errors or omissions.
- (b) If a candidate response does not seem to be covered by either the principles or detailed marking instructions, and you are uncertain how to assess it, you must seek guidance from your team leader.
- (c) Do not award half marks.
- (d) Where a candidate makes an error in the first part of a question, award marks for subsequent answers that are correct with regard to this original error. Do not penalise candidates more than once for the same error.
- (e) Unless a numerical question specifically requires evidence of working to be shown, award full marks for a correct final answer (including units, if appropriate) on its own.
- (f) Candidates should not use bulleted lists to answer extended-response questions. They must respond to the 'command' word as appropriate and provide extended answers to communicate fully their knowledge and understanding. Candidate responses in the form of bulleted lists may not be able to access the full range of available marks.
- (g) In the detailed marking instructions, if a word is underlined then it is essential; if a word is **(bracketed)** then it is not essential.
- (h) In the detailed marking instructions, words separated by / are **alternatives**.
- (i) A correct response can be negated if the candidate includes:
 - an extra, incorrect, response
 - additional information that contradicts the correct response
- (j) Where the candidate is instructed to choose one question to answer but instead answers two questions, mark both responses and award the higher mark.
- (k) Unless otherwise required by the question, the use of abbreviations (for example DNA, ATP) or chemical formulae (for example CO₂, H₂O) are acceptable alternatives to naming.
- (l) If a numerical answer is required and units are not given in the stem of the question or in the answer space, candidates must supply the units to gain the mark. If units are required on more than one occasion, do not penalise candidates repeatedly.
- (m) If incorrect spelling is given:
 - If the correct word is recognisable then award the mark.
 - If the word can easily be confused with another biological term then **do not** award the mark, for example glucagon and glycogen.

(n) Presentation of data:

- If a candidate provides two graphs, in response to one question, mark both and award the higher mark.
- If a question asks for a particular type of graph/chart and the candidate gives the wrong type, do not award full marks. Candidates cannot achieve the plot mark but **may** be able to achieve the mark for scale and label. If the x and y data are transposed, then do not award the scale and label mark.
- If the graph uses less than 50% of the axes then do not award the scale and label mark.
- If 0 is plotted when no data for this is given, then do not award the plot mark – candidates should only plot the data given.

(o) Only award marks for a valid response to the question asked. For example, in response to questions that ask candidates to:

- **identify, name, give or state**, they need only answer or present in brief form
- **describe**, they must provide a statement as opposed to simply one word
- **explain, justify**, they must provide a reason for the information given
- **compare**, they must demonstrate knowledge and understanding of the similarities and/or differences between topics being examined
- **calculate**, they must determine a number from given facts, figures or information
- **predict**, they must indicate what may happen based on available information
- **suggest**, they must apply their knowledge and understanding to a new situation

2024 Higher Biology Section 1 Multiple Choice Analysis

Item	Key	A	B	C	D	Missing	Invalid
1	D	3	15	5	78	0	0
2	B	4	81	6	8	0	0
3	D	15	8	2	76	0	0
4	D	4	3	5	89	0	0
5	B	8	73	2	17	0	0
6	C	23	8	65	4	0	0
7	B	7	63	11	20	0	0
8	B	22	50	17	11	0	0
9	A	67	7	23	2	0	0
10	D	20	5	4	70	0	0
11	B	29	53	17	1	0	0
12	B	28	55	9	8	0	0
13	A	76	8	10	5	0	0
14	C	3	8	80	9	0	0
15	C	9	31	56	4	0	0
16	B	9	71	6	15	0	0
17	D	2	4	8	85	0	0
18	A	84	8	3	5	0	0
19	C	16	15	52	16	0	0
20	C	24	8	66	1	0	1
21	C	48	6	33	13	0	0
22	A	43	34	13	9	0	0
23	D	20	2	4	73	0	0
24	A	56	5	7	32	0	0
25	C	12	29	47	12	0	0

2024 Higher Biology Section 2 Marks Data Analysis

2024 Item	Max Mark	Average Mark	% Average Mark	Discrimination	% Not Attempted
Q1a	2	1.5	72.7%	57	1.0%
Q1bi	1	0.8	78.5%	41	1.5%
Q1bii	1	0.7	69.3%	57	4.8%
Q1ci	1	0.8	77.0%	21	2.0%
Q1cii	2	0.4	21.2%	44	4.6%
Q2a	1	0.5	45.1%	45	4.6%
Q2bi	2	1.7	87.4%	45	0.8%
Q2bii	1	0.1	10.9%	29	3.2%
Q2ci	1	0.3	29.5%	52	7.3%
Q2cii	2	0.4	21.1%	55	5.5%
Q3ai	1	0.6	60.8%	55	8.3%
Q3aii	2	1	48.2%	58	5.1%
Q3bi	1	0.3	33.2%	43	13.2%
Q3bii	1	0.7	69.7%	33	2.2%
Q4a	1	0.7	71.8%	49	4.3%
Q4bi	1	0.7	69.5%	52	5.1%
Q4bii	1	0.2	16.5%	41	7.9%
Q4c	2	1.4	68.7%	45	3.9%
Q5a	1	0.6	64.5%	33	0.7%
Q5bi	1	0.7	74.1%	39	1.4%
Q5bii	1	0.7	70.7%	25	1.6%
Q5c	2	0.5	25.4%	45	6.1%
Q5d	2	1.1	56.8%	49	2.5%
Q6ai	1	0.6	61.4%	35	0.7%
Q6aii	1	0.7	71.1%	46	1.4%
Q6bi	1	0.3	33.4%	38	1.3%
Q6bii	2	1	49.9%	51	4.0%
Q6biii	1	0.6	56.4%	37	2.2%
Q7a	1	0.7	70.0%	33	0.5%
Q7bi	2	1.4	68.0%	51	1.1%
Q7bii	1	0.6	57.5%	49	1.7%
Q7c	1	0.5	49.9%	33	2.1%
Q7d	1	0.7	71.2%	35	3.1%
Q8ai	1	0.6	63.2%	58	11.4%
Q8aii	1	0.3	26.0%	41	16.5%
Q8b	1	0.6	59.6%	60	7.8%
Q8c	2	0.5	25.3%	57	30.4%
Q9A	4	2.8	69.7%	68	2.6%
Q9B	4	2.7	67.8%	68	0.0%
Q10ai	1	0.2	19.1%	28	1.1%
Q10aii	1	0.7	65.6%	31	0.8%
Q10b	2	1.6	79.5%	32	3.0%
Q10c	1	0.4	37.1%	42	2.5%
Q10d	2	0.5	22.6%	44	6.4%
Q11ai	2	0.6	32.0%	48	2.6%
Q11aii	1	0.7	70.8%	47	3.3%
Q11aiii	1	0.6	61.1%	51	2.3%
Q11aiv	1	0.4	44.6%	63	12.7%
Q11bi	1	0.5	46.4%	31	4.2%
Q11bii	1	0.2	22.9%	33	6.8%
Q11ci	1	0.5	50.3%	43	7.6%
Q11cii	1	0.3	34.4%	56	6.2%
Q12a	1	0.4	40.1%	58	7.8%
Q12b	1	0.2	15.2%	44	1.7%
Q12c	1	0.4	40.8%	43	2.8%
Q12di	1	0.8	76.7%	40	1.9%
Q12dii	2	1.2	58.6%	20	1.4%
Q13ai	1	0.4	40.7%	33	2.6%
Q13aii	1	0.8	77.6%	30	1.2%
Q13b	1	0.2	24.0%	37	2.4%
Q13c	1	0.6	62.8%	55	2.9%
Q13d	1	0.9	88.3%	27	1.5%
Q13e	2	0.5	23.6%	56	4.9%
Q14ai	1	0.4	38.4%	40	4.8%
Q14aii	1	0.7	66.8%	33	1.0%
Q14bi	2	0.8	39.6%	52	2.5%
Q14bii	1	0.6	60.3%	54	4.6%
Q14biii	1	0.2	17.7%	38	4.2%
Q15A	8	4.5	55.9%	72	4.9%
Q15B	8	5.1	63.1%	71	0.0%
Paper	95	49	51.6%		

2024 Higher Biology Section 2 Distribution of Marks Analysis

2024 Item	0	1	2	3	4	5	6	7	8
Q1a	15	24	61						
Q1bi	21	79							
Q1bii	31	69							
Q1ci	23	77							
Q1cii	72	13	15						
Q2a	55	45							
Q2bi	3	19	78						
Q2bii	89	11							
Q2ci	71	29							
Q2cii	71	15	14						
Q3ai	39	61							
Q3aai	35	34	31						
Q3bi	67	33							
Q3bii	30	70							
Q4a	28	72							
Q4bi	31	69							
Q4bii	84	16							
Q4c	18	27	55						
Q5a	35	65							
Q5bi	26	74							
Q5bii	29	71							
Q5c	61	27	12						
Q5d	22	41	36						
Q6ai	39	61							
Q6aai	29	71							
Q6bi	67	33							
Q6bii	39	22	39						
Q6biii	44	56							
Q7a	30	70							
Q7bi	17	30	53						
Q7bii	43	57							
Q7c	50	50							
Q7d	29	71							
Q8ai	37	63							
Q8aai	74	26							
Q8b	40	60							
Q8c	69	11	20						
Q9A	8	10	17	25	40				
Q9B	9	15	15	17	43				
Q10ai	81	19							
Q10aai	34	66							
Q10b	10	22	69						
Q10c	63	37							
Q10d	63	30	8						
Q11ai	49	39	13						
Q11aai	29	71							
Q11aaii	39	61							
Q11aaiii	55	45							
Q11bi	54	46							
Q11bii	77	23							
Q11ci	50	50							
Q11cii	66	34							
Q12a	60	40							
Q12b	85	15							
Q12c	59	41							
Q12di	23	77							
Q12dii	33	18	50						
Q13ai	59	41							
Q13aai	22	78							
Q13b	76	24							
Q13c	37	63							
Q13d	12	88							
Q13e	58	38	5						
Q14ai	62	38							
Q14aai	33	67							
Q14bi	43	36	22						
Q14bii	40	60							
Q14biii	82	18							
Q15A	12	7	7	9	11	12	13	12	17
Q15B	5	5	7	11	12	13	13	12	22

2024 Higher Biology Section 3 Assignment Marks Analysis

Item	Max Mark	Average Mark	% Average Mark	Discrimination
1	1	0.8	81.9%	28
2	4	2.4	58.8%	44
3a	1	0.5	52.6%	32
3b	1	1	97.6%	12
3c	1	0.5	48.2%	37
3d	1	0.6	58.4%	26
3e	1	0.5	46.2%	39
4a	1	1	96.4%	18
4b	1	0.8	81.1%	22
4c	1	0.9	87.2%	25
4d	1	0.7	71.7%	31
5	1	0.3	25.8%	40
6	1	0.4	40.6%	34
7	3	1.2	39.0%	49
8	1	0.8	78.9%	22
Paper	20	12.2	61.0%	

Item	Max Mark	0	1	2	3	4
1	1	18	82			
2	4	12	17	22	20	29
3a	1	47	53			
3b	1	2	98			
3c	1	52	48			
3d	1	42	58			
3e	1	54	46			
4a	1	4	96			
4b	1	19	81			
4c	1	13	87			
4d	1	28	72			
5	1	74	26			
6	1	59	41			
7	3	32	33	23	13	
8	1	21	79			
Paper	20					

Item	Max Mark	Item Description
1	1	An aim that describes clearly the purpose of the investigation
2	4	Underlying biology
3a	1	A brief summary of the approach used to collect experimental data.
3b	1	Sufficient raw data from the candidate's experiment.
3c	1	Data, including mean/average values, presented in a correctly produced table(s)
3d	1	Data relevant to the aim from an internet/literature source
3e	1	A citation and reference for a source of the internet/literature data
4a	1	An appropriate format from the options of bar graph or line graph.
4b	1	The axes of the graph have suitable scales.
4c	1	The axes of the graph have suitable labels and units
4d	1	Accurately plotted data points and a line (line graph) or clear bar tops (bar graph)
5	1	A valid comparison or an appropriate calculation linked to the aim of the investigation.
6	1	A valid conclusion that relates to the aim and is supported by all the data in the report.
7	3	Evaluation of the investigation
8	1	A clear and concise report, with an informative title.